

Chapter 1, Problem 4PP

Problem

To move charge q from point b to point a requires 25 J. Find the voltage drop v_{ab} (the voltage at a positive with respect to b) if: (a) $q = 5$ C, (b) $q = -10$ C.

Step-by-step solution

Step 1 of 2

(a)

The energy required to move 5 C charge from point b to point a is 25 J.

Calculate the voltage drop.

$$\begin{aligned}v_{ab} &= \frac{25 \text{ J}}{5 \text{ C}} \\&= 5 \text{ V}\end{aligned}$$

Hence, the voltage drop between points a and b is $\boxed{5 \text{ V}}$.

Step 2 of 2

(b)

The energy required to move -10 C charge from point b to point a is 25 J.

Calculate the voltage drop.

$$\begin{aligned}v_{ab} &= \frac{25 \text{ J}}{-10 \text{ C}} \\&= -2.5 \text{ V}\end{aligned}$$

Hence, the voltage drop between points a and b is $\boxed{-2.5 \text{ V}}$.